NAVI 20% Review: 15-Minute Speaker Notes

OCT 12, 2026 · SAYANJEET DAS

The deck fits 15 minutes: talk time is trimmed to 14:15, so six handoffs of about 5 seconds land you near 14:45 with 15 seconds spare. The opening is split across three speakers.

Run sheet

Seven blocks, six handoffs. Keep a stopwatch running; if you are past the "Ends by" time, cut to your last slide's one-line takeaway.

#	SPEAKER	SLIDES	TALK	STARTS	ENDS BY
1	Sayanjeet Das	1–2 · opening, problem	1:10	0:00	1:10
2	Carollo Aradhan V	3–4 · existing approaches, objectives	1:35	1:15	2:50
3	Utkarsh D. Ghorpade	5–7 · literature	2:15	2:55	5:10
4	N Arun	8–10 · literature, concept, architecture	2:35	5:15	7:50
5	Carollo Aradhan V	11–12 · hardware, exploded model	1:55	7:55	9:50
6	Muhammed Farhan	13–15 · algorithms, software, coverage	2:50	9:55	12:45
7	Sayanjeet Das	16–18 · milestone, plan, conclusion	1:55	12:50	14:45

Totals: Sayanjeet 3:05 · Carollo 3:30 · Utkarsh 2:15 · Arun 2:35 · Farhan 2:50. Sayanjeet opens and closes.

Rehearsal rules

Do two full timed run-throughs; the first usually runs 1–2 minutes long.

1. Install the fonts from the NAVI_Fonts folder on the presenting laptop, then open the deck in Slide Show once to check the animations.
2. One person drives the clicker for the whole talk, so speakers never fumble keys. Say "next" or nod for each click.
3. Literature slides 5–8 are the easiest to overrun: say the one finding and the NAVI decision, never read the table.
4. On slides with many clicks (10, 15), keep talking while the next element appears; don't wait for it to finish.
5. Handoff = one sentence naming the next speaker and topic, then step aside. No thank-yous between speakers.
6. If you are 30 s behind at a handoff, the next speaker drops their last bullet. Never skip a slide.
7. Backups A09–A21 are for Q&A only: type the slide number + Enter to jump (A-number + 18, e.g. A15 = 33).

Sayanjeet Das — slides 1–2 (1:10) and 16–18 (1:55)

OPENING · STARTS AT 0:00

Slide 1 · Title (0:30)

no clicks

- "We are presenting NAVI, the Navigation Assistant for Visitor Interaction: a GPS quadcopter that *leads* a walking visitor along mapped, inspected campus routes."
- "This is our 20% review: what is designed, computed and simulated, and what still needs physical testing."
- Name the five members in one breath; point to the signature box only if the panel asks.

Slide 2 · The problem (0:40)

2 clicks

- Click 1 (journey): "Every aid today asks the visitor to first find themselves on a map, then turn it into turns. Step 03, self-localising, is the friction."
- Click 2 (cards): "Our users are first-time visitors, mostly parents and guests. The campus is 42.2 hectares; the median route from the gate is 453 m."
- Do not say visitors "always get lost" — we have not measured that.
- Handoff: "Carollo will take you through existing approaches and our objectives."

CLOSING · STARTS AT 12:50

Slide 16 · The 20% milestone (0:55)

3 clicks

- Click 1 (chart): "These are computed graph distances. From the main gate, only 12 of 36 destinations fit in one 350 m leg; from the nearest pad, every leg is at most 316 m."
- "The zero for E Block means the pad and entrance share a graph node — not landing accuracy."
- Click 2: done = graph, coverage, apps in simulation, design model; pending = timed hover, phone GPS error, obstruction survey.
- Click 3: "20% is our milestone label, not an audited percentage."

Slide 17 · Plan change & gates (0:35)

1 click

- "The lab's S500, Pixhawk and RTK kit was not available, so we chose a self-funded F450 that still runs ArduPilot and MAVLink."
- "Budget pool is ₹25,000; our midpoint estimate is about ₹23,500, but the range reaches ₹29,900, so we buy in stages."
- Click 1: name Phases 0–2 only: permission and purchase, bench tests, timed hover.

Slide 18 · Conclusion (0:25)

1 click

- "At 20% we have the route graph, coverage, simulated software and a design model. Nothing has flown yet."
- "Next evidence comes from bench failsafes and a timed hover. Thank you — we welcome your questions."

Carollo Aradhan V — slides 3–4 (1:35) and 11–12 (1:55)

FIRST BLOCK · STARTS AT 1:15

Slide 3 · Existing wayfinding (0:40)

2 clicks

- Click 1: "Signs are passive, escorts don't scale on event days, phone maps stop at the gate, and the official VIT wayfinder is excellent but still a map."
- Click 2: "The wayfinder's own path data — 278 waypoints, 36 entrances — is our data source. NAVI complements it by adding a physical guide."

Slide 4 · Problem statement & objectives (0:55)

3 clicks

- Read the problem statement aloud once, slowly.
- Clicks 1–2: run the five objectives in one line each; stress that every "Verify by" is a *proposed* acceptance target.
- Click 3: "In scope: outdoor, inspected routes, one visitor, daytime, supervised. Out of scope: indoor, night, crowds, and live obstacle detection."
- Handoff: "Utkarsh will take the research behind NAVI."

SECOND BLOCK · STARTS AT 7:55

Slide 11 · The F450 platform (0:55)

3 clicks

- Opening: "This is our CAD design model with nominal dimensions — not a photograph and not built yet."
- Click 1 (flight control): "SpeedyBee F405 V3, bought as the FC board only, because the F450 kit already has four separate 30 A ESCs. The vendor photo shows an extra 4-in-1 ESC we are not buying."
- Click 2 (propulsion): "A2212 1000KV motors, 10-inch props, and a 3S 3300 mAh battery with an XT60 plug."
- Click 3 (positioning): "M8N GPS with compass on a mast, SiK 433 MHz radio, and a power module. The transmitter, charger and safe bag stay on the ground."

Slide 12 · Exploded design model (1:00)

2 clicks

- Before clicking: "Here is the assembled model."
- Click 1 (parts separate): "Each layer is a real component group from our CAD, moved by the model's own explode offsets."
- Click 2 (labels): walk top to bottom: props, motors, GPS mast, avionics, plates, arms, ESCs, battery, landing gear. Point at the inset for the small avionics, items 3–7.
- Close with the caveat on screen: "Nominal geometry; physical assembly and measurement are pending."
- Handoff: "Farhan will cover the software and results."

Utkarsh Dilipkumar Ghorpade — slides 5–7 (2:15) · starts at 2:55

Slide 5 · Literature 1: human–drone interaction (0:45)

`1 click`

- "Our recent window is 2021–2026; cyan tags are recent, grey are foundational."
- "DroneNavigator (2017) guided 14 visually impaired users with a drone; it was faster than audio, but indoors and at 1 m."
- "Wögerbauer 2024 found people prefer 1.6–1.9 m from a virtual drone. That does **not** validate our 15 m lead — the 15 m comes from GPS error."

Slide 6 · Literature 2: wayfinding (0:45)

`2 clicks`

- Click 1 (rows): "Kuys 2024 showed landmarks help people learn a route; Selvi tested a campus app with 14 first-time visitors."
- "OSM landmark work only holds where map data is complete — so we use OSM only for the campus boundary."
- Click 2 (cyan column): read just the NAVI decisions, not the whole table.

Slide 7 · Literature 3: positioning (0:45)

`3 clicks`

- Click 1 (top strip): "Aircraft GPS is about 2.5–5 m; a phone is about 7–13 m. Both are estimates, not campus measurements."
- Click 2: "Indoor UWB reached 0.12 m, but indoors with anchors — we don't claim that accuracy. The HMM map-matching paper motivates snapping, but we have not built an HMM."
- Click 3: "A browser can't give raw phone sensors, so we will measure handset error instead."
- Handoff: "Arun takes planning research and our system design."

N Arun — slides 8–10 (2:35) · starts at 5:15

Slide 8 · Literature 4: planning & evaluation (0:45)

2 clicks

- Click 1: "The 2025 obstacle-avoidance survey shows this is a whole field; we have no live obstacle sensing, so routes are inspected instead."
- "The path-planning survey says no planner is best everywhere; on a known, static graph, Dijkstra is the right tool."
- "Two papers were only partly accessible, and the slide says so."
- Click 2: read the cyan column quickly.

Slide 9 · Concept of operations (0:45)

3 clicks

- Click 1 (route draws): "The visitor picks a destination; we route from the nearest pad, check the geofence and the waypoint whitelist, upload the mission, and the aircraft flies about 15 m ahead."
- Click 2: "If the visitor stops, the aircraft holds."
- Click 3: "We don't claim waypoint flight or aerial guidance is new. Our work is combining routes, pad coverage, phone-position pacing and fault responses on affordable hardware."

Slide 10 · Architecture (1:05)

4 clicks

- Opening: "All planning runs on the ground station. The F405 runs ArduPilot for stabilisation and onboard failsafes. There is no onboard computer."
- Click 1: phone or kiosk sends destination and position over HTTPS.
- Click 2: ground station sends the mission and pacing commands over MAVLink on the 433 MHz SiK radio.
- Click 3: aircraft position and status come back to the live map.
- Click 4: "The RC radio is a separate link, so the safety operator can always take over."
- Say clearly: "Telemetry is simulated today; the MAVLink link is pending."
- Handoff: "Carollo will show the hardware."

Muhammed Farhan — slides 13–15 (2:50) · starts at 9:55

Slide 13 · Routing and pacing modules (0:55)

2 clicks

- Click 1, left side: "Haversine distances, Dijkstra from the pad to the destination's nearest entrance, leg splitting at 350 m, greedy pad coverage, and a geofence with a 40 m buffer. These run in our app's simulation."
- Right side: "Pacing is proposed. We project the visitor onto the route and compare progress, not straight-line distance. Hysteresis means the aircraft doesn't flicker between lead and hold. A position older than 30 s means hold, then return to launch."
- Click 2: "Precedence: failsafe, then geofence, then stale data, then pacing."

Slide 14 · Software evidence (1:00)

2 clicks

- "These are our app screens. Note the status chips: NO FIX and SIMULATION. The route panel shows Gazebo from South Pad, 157 m, 3 min 16 s — that's 0.8 m/s walking speed."
- "The code panel is labelled pseudocode — it shows the proposed logic, not a live controller."
- Click 1: "We inject a fault: visitor position stale for 30 seconds."
- Click 2: "The designed response steps SAFE, WARNING, HOLD, RETURN TO LAUNCH. Firmware and field tests are pending."

Slide 15 · Coverage results (0:55)

4 clicks

- Click 1 (map): the red dashed line is the OSM campus boundary.
- Click 2 (pads): "Six pads from a greedy set-cover; pad 6, E Block, sits under the route panel in this screenshot."
- Click 3 (route): one computed route.
- Click 4 (numbers): "260 flight nodes, 303 segments, 8.24 km, 36 destinations in one connected graph; the longest nearest-pad leg is 316 m."
- Say: "Greedy, not proven minimal, and computed, not flown. Coverage gets recomputed once endurance is measured."
- Handoff: "Sayanjeet will close with the milestone and plan."

Q&A: who answers what

Answer pattern: state the evidence, name what is uncertain, then the test that will settle it. Never present a planned test as a result. Backup slides A09–A21 hold the detail.

LIKELY QUESTION	ANSWERS	ONE-LINE ANSWER	BACKUP
What does the 20% actually show?	**Sayanjeet**	Computed graph and coverage, simulated apps with fault injection, platform choice and CAD. No hardware, no flight.	—
How does NAVI differ from the VIT wayfinder?	**Sayanjeet**	The wayfinder's data is our source; NAVI removes the self-localising step any map leaves.	—
Waypoint flight exists — what's new?	**Arun**	Pacing an aircraft to a walking person, plus pads, geofence and zero-install guidance.	—
What if the phone locks or the link drops?	**Arun**	Stale over 30 s means hold then return; telemetry loss means return or land; RC operator stays ready.	**A15**
Why no obstacle detection?	**Arun**	Out of scope and budget; routes are inspected, altitude surveyed per segment, RC takeover.	**A15**
Why the F450 and only the F405 FC?	**Carollo**	Cheapest platform running ArduPilot and MAVLink; the kit already has four ESCs.	**A18**
Is it built? What's left to measure?	**Carollo**	No. Timed hover, battery capacity, phone GPS error, obstructions, GPS spot-checks, wind.	**A17**
Why do photos show 915 MHz or an extra ESC?	**Utkarsh**	They are vendor reference photos; we plan 433 MHz, XT60, FC only. Flagged, not edited.	**A10**
Is 350 m endurance validated?	**Utkarsh**	No. The hover estimate is unmeasured; a timed hover to failsafe will reset the limit.	**A14**
Does a 15 m lead absorb GPS error?	**Utkarsh**	Not provably; the error estimates could exceed it. We'll measure handsets, then set thresholds.	**A13**
What could change the six-pad result?	**Farhan**	Measured endurance, walking speed, reserve, and pad site clearance. It's greedy, not minimal.	**A14**
What if the drone isn't ready?	**Farhan**	Our ground robot runs the same navigation code as a backup demo.	—